

MA 545: Technical Analysis of Financial Markets

Course Code: MA 545

Course Name: Technical Analysis of Financial Markets

L-T-P-C: 2-0-3-4

Prerequisites: IC 252, IC 272 or equivalent

Intended for: UG, PG, PhD

Mutual Exclusion: NA

Distribution: Elective

Approval: 65th BoA (replaces MA 654)

Preamble

This course introduces students to Technical Analysis for investment decision making through the study of past market data. It complements the fundamental and quantitative approaches to investing by focusing on market behaviour and chart interpretation. Beginning with the philosophical foundations and core assumptions of technical analysis, the course develops the major analytical frameworks used by practitioners, spanning trend and price-structure analysis, candlestick and chart-pattern analysis, and technical indicators. Emphasis is also placed on the designing rules-based trading systems and backtest them using charting platforms and programming tools. By the end of the course, students will be able to read and interpret financial charts, apply and evaluate the principal technical frameworks, build and test trading strategies, and critically assess their validity. The laboratory sessions ensure that concepts are applied to financial market data through hands-on charting and simulation.

Course Modules with Quantitative Lecture Hours

- **Unit/Topic 1: Foundations of Technical Analysis:** Philosophy and core assumptions of technical analysis. Technical versus fundamental approaches, and Efficient Market Hypothesis. Basics of price and volume, and the principal chart types and timeframes. (2 Lectures)
- **Unit/Topic 2: Trend and chart pattern analysis:** Dow Theory as a foundational framework. The concept of trend and its classification. Trendlines and channels, support and resistance, and other structural features of price and their interpretation. Japanese candlestick analysis and the interpretation of common formations. Classical reversal and continuation chart patterns, and the principles of pattern recognition and price projection. (6 Lectures)
- **Unit/Topic 3: Technical Indicators:** Trend-following indicators, Momentum oscillators and the concept of divergence. Volume- and volatility-based indicators, and the construction and combination of indicators. (4 Lectures)
- **Unit/Topic 4: Trading Systems, Risk Management and Backtesting:** Design of rules-based trading systems. Principles of risk and money management. Backtesting methodology, In-sample and out-of-sample testing, rolling-window evaluation based on return, volatility, Sharpe ratio, maximum drawdown, and turnover, performance evaluation, and common pitfalls: look-ahead bias, survivorship bias, data snooping, multiple testing, and overfitting. Transaction costs, slippage, and market impact on the performance. (8 Lectures)
- **Unit/Topic 5: Machine Learning Applications in Technical Analysis:** Machine learning as an extension of rule- and indicator-based technical analysis. Feature construction from market data and the

formulation of trading problems as learning tasks. Representative supervised learning methods for signal generation, and their training, validation, and evaluation on financial data. Critical discussion of the pitfalls specific to financial machine learning. (8 Lectures)

Laboratory/Practical/Tutorial Modules

The course includes a weekly laboratory in which students apply the concepts studied to real market data using charting platforms and programming tools. The laboratory work progresses from reading and annotating charts, through the construction of indicators and patterns, to the design and backtesting of trading strategies with appropriate risk management, and further to the application of machine learning methods to market data.

Text Books

1. Pring, M. J., *Technical Analysis Explained*. McGraw Hill, 5th Edition, 2014.
2. Chan, E. P., *Algorithmic Trading: Winning Strategies and Their Rationale*. Wiley, 2013.

References

1. Murphy, J. J., *Technical Analysis of the Financial Markets*. New York Institute of Finance / Penguin, 1999.
2. Nison, S., *Japanese Candlestick Charting Techniques*. Prentice Hall Press, 2nd Edition, 2001.
3. Kaufman, P. J., *Trading Systems and Methods*. John Wiley & Sons, 2019.